

OPTIMALITY THEORETIC SYNTAX: PRESENTATION

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Optimality Theory (OT) is a generative theory that differs substantially from the generative ‘derivational’ tradition by emphasizing the representational character of grammatical constraints (cf. Prince & Smolensky 1993, McCarthy & Prince 1993). Much of the work in OT has been developed in the areas of Phonology and Morphology, and these areas have already presented several interesting published and unpublished papers, articles, books and dissertations. Studies in OT Syntax, on the other hand, were not so numerous, but by the end of the 1990s, they had already become a viable alternative to the derivational approaches (cf. Barbosa et al. 1998, Legendre Grimshaw & Vikner 2001, Sells 2001). However, unlike the productive fields of OT Phonology and OT Morphology, there are not many researchers applying an OT approach to the study of Syntax in a systematic way.

Nevertheless, we believe that optimality-theoretic syntax offers a new perspective on old issues in syntactic theory. A major innovation of OT is that it is essentially a formalization of the idea that grammar is a tool for resolving conflicts among principles constraining the well-formedness of language expressions. This concept requires that grammatical principles be evaluated simultaneously, in parallel – without that, there is no conflict among constraints to begin with. A natural consequence is that it is not necessary to conceive the relationship among different components of grammar – Lexicon, Morphology, Syntax, Semantics, Prosody – as being ‘linear’, ‘sequential’; this relationship may simply be ‘parallel.’ In other words, conditions from different grammatical components may, in OT syntax, operate simultaneously and get in conflict with each other.

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Of course, the idea of a ‘parallel’ model of grammar was revolutionary within the Chomskyan generative linguistics. In this tradition, the architecture of grammar, at least as far as Syntax was concerned, was always derivational, with well-formedness being the result of the successive application of operations and verification of their output. On the other hand, it should be recalled that other branches of generative linguistics were already developing non-derivational approaches by the middle of the seventies (as in the case of GPSG and LFG). But, for people working within the Chomskyan tradition, OT opened new venues for approaching the complexity of syntax – especially by emphasizing complexity as an apparent result of interaction rather than ‘true complexity’ of rule application or output representation. For example, one of the first treatments of syntactic phenomena in OT was Jane Grimshaw and Vieri Samek-Lodovici’s 1995 paper on optional subjects. This paper ‘officially’ incorporated, for the first time, a pragmatic constraint on null subjects into the analysis of the phenomenon, allowing for simplification of the syntactic mechanisms required (in particular, the concept of ‘small pro’). Although similar in spirit, Chomsky’s ‘Avoid Pronoun Principle’, precisely because it was never really an ‘official’ part of the theory, did not lead to the same conclusion.

In this special issue of ReVEL, on OT Syntax, the reader will find a selection of articles that discuss or adopt the approach we have just referred to². We gathered well-known researchers working with OT Syntax in order to give a new overview of the ideas being developed in the field today.

Ellen Woolford presents an analysis of agreement in Choctaw and Lakota. Agreement in these languages is expressed both by a system of pronominal clitics and by what the author calls ‘T agreement’. Woolford’s approach stresses the interactional character of the phenomenon of agreement: while ‘syntax is crucially involved, limiting the range of [subject] agreement, the main action is at the syntax/PF interface, where constraints on morphological spell-out interleaved with phonological constraints determine whether T agreement or a pronominal clitic will be used to cross-reference a nominative argument.’

Géraldine Legendre and Mary Schindler’s paper brings ‘evidence that Urban Wolof (UW) fits nicely into the cross-linguistic code-switching typology, once syntactic constraints on code-switching are construed as violable, as proposed in Optimality Theory’. The authors demonstrate that ‘UW permits DPs consisting of a French lexical noun immediately followed by a Wolof determiner as well as French lexical verbs immediately followed or preceded by

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Wolof inflection’. On the basis of this sort of evidence, the authors argue ‘against an alternative Minimalism-based approach which places code-switching at the syntax-PF interface’.

In his paper, Luigi Burzio returns to a set of phenomena to which he has dedicated many articles – patterns of anaphoric relations and, specially, of correlations between (un)availability of types of pronouns and reflexives. The paper offers a deep synthesis of many empirical generalizations, aiming at establishing a particular correlation: the same antecedents that fail in long-distance relations also require strong anaphors locally. According to Burzio, a constraint interaction approach to this correlation is able to derive an important result of the eighties’ literature on chain formation, namely, what Burzio calls ‘the effects of Rizzi’s Chain Condition’: the fact that reflexive clitics are banned, in languages like French and Italian, with syntactically derived subjects. (Beware that ‘Chain Condition’ here is actually a theorem in Rizzi’s representational approach to chain formation, and has nothing to do with Reinhart & Reuland 1993’s Chain Condition.) The upshot of Burzio’s paper – as in most classical OT papers – is that both wide range generalizations and fine-grained construction-specific properties can be simultaneously derived from interaction of independent, unrelated constraints. Hence, the complexity of grammatical phenomena do not need to be build into overarching principles of grammar.

Geoffrey Poole and Noel Burton-Roberts discuss the notion of correspondence between levels in grammar. The vagueness of the traditional notion of ‘correspondence’ – and in particular the implicit equation of ‘correspondence’ and ‘resemblance’ – leads to a view of the lexicon and of the sound/meaning relation which, according to the authors, is problematic. They argue in favor of a different kind of relation between levels, based on the notion of ‘m-representation’: ‘M-representation is in part motivated by the sortal distinctness between the content of phonological features and the content of semantic features’. They argue, moreover, that ‘this concept finds a natural home within the architecture of OT’.

Rodrigo Gutiérrez-Bravo focuses on the nature of the input in Optimality Theoretic syntax. Recent works (Kuhn 2003, Beaver & Lee 2004, Heck et al. 2002) have argued that, unlike the standard assumption in OT phonology, OT syntax can eliminate faithfulness constraints, the input, or both. In his paper, Gutiérrez-Bravo ‘challenges this view and argues in favor of a fully structured input and constraints that target the feature content of the input’. His proposal is supported by a study of subjects and topics in Spanish infinitival clauses.

Hans Broekhuis & Wim Klooster address what seems to be an issue closer to minimalist inquiries, namely, the question whether Merge is inherently more economical than ReVEL, special edition n. 4, 2010.

Move, blocking the latter (as in Chomsky 1995). They investigate negative sentences in Dutch and English, on the basis of which they conclude that both Merge and Move are equally costly operations; therefore, the computation cannot be based on a numeration; instead, it needs to have free access to the lexicon. For OT readers, the main interest of the paper lies in the authors' attempt to conciliate minimalist and OT ideas in a model called Derivation-and-Evaluation, previously developed by Broekhuis and Dekkers (2000) and Broekhuis (2000).

Like Broekhuis & Klooster's paper, Christoph Gabriel's also ends up arguing for an approach that combines Minimalism and OT Syntax. It presents a focused-conditioned account to explain word order phenomena in two varieties of Argentinean Spanish. As Christoph Gabriel points out, '[f]ormal approaches that integrate pragmatically motivated features correctly predict instances of focus-induced word order, but run into problems when a given focus-background articulation corresponds to different possible output forms, hence when optionality is at play.' And, of course, this is also problematic for the standard ranking of classical OT. The paper argues that the different syntactic and phonological strategies of focus marking in the two varieties of Argentinean Spanish studied can be explained by a combination of Chomsky's (2000, 2001) target/probe approach and Stochastic OT (Boersma & Hayes 2001).

Lestrade, de Schepper, Westelaken & de Hoop's paper focuses on the discussion of the 'R-form of scrambled pronominal P complements in Dutch.' The authors analyze the phenomenon within the OT framework, proposing an explanation to '[t]he fact that pronominal P complements that do not refer to humans scramble, and subsequently take up a spatial form' using five constraints: ECONOMY and STAY (keep the basic word order of the language), SCRAMBLE (pronominal constituents must appear on the beginning of the sentence), PCASE (mark the syntactic relationship between the preposition and its complement), and *RHUM (the combination of a human referent with a spatial form is forbidden).

REFERENCES

1. BARBOSA, P. et al. (eds.) *Is the best good enough? Optimality and competition in syntax*. Cambridge: MIT Press, 1998.

2. BEAVER, D. & H. LEE. Input-Output mismatches in OT. In R. Blutner & H. Zeevat (Eds.). *Optimality Theory and Pragmatics*, 112-153. New York: Palgrave Macmillan, 2004.
3. BOERSMA, Paul; HAYES, Bruce. Empirical Tests of the Gradual Learning Algorithm. *Linguistic Inquiry* 32, 2001.
4. BROEKHUIS, Hans. Against feature strength: the case of Scandinavian object shift. *Natural Language and Linguistic Theory* 18:673-721, 2000.
5. BROEKHUIS, Hans; DEKKERS, Joost. The minimalist program and optimality theory: derivations and evaluations. In: DEKKERS, Joost et al. (eds.) *Optimality Theory: phonology, syntax and acquisition*. Oxford/New York, Oxford University Press, 2000.
6. CHOMSKY, Noam. *The minimalist program*. Cambridge (Mass.): MIT Press, 1995.
7. CHOMSKY, Noam. Minimalist Inquiries: the Framework. In: MARTIN, Roger; MICHAELS, David; URIAGEREKA, Juan. *Step by step: Essays on minimalist syntax in honor of Howard Lasnik*. Cambridge, Mass.: MIT Press, 2000.
8. CHOMSKY, Noam. Derivation by phase. In: KENSTOWICZ, Michael. *Ken Hale: A Life in Language*. Cambridge, Mass.: MIT Press, 2001.
9. GRIMSHAW, G.; SAMEK-LODOVICI, V. Optimal subjects. In: BECKMAN, J. et al. (eds.) *University of Massachusetts occasional papers 18: papers in Optimality Theory*. GSLA, University of Massachusetts, Amherst, 1995.
10. HECK, F., G. MÜLLER, R. VOGEL, S. FISCHER, S. VIKNER & T. SCHMID. On the nature of the input in Optimality Theory. *The Linguistic Review* 19:345-376, 2002.
11. KUHN, J. *Optimality Theoretic Syntax: A declarative Approach*. Stanford: CSLI, 2003.
12. LEGENDRE, G.; GRIMSHAW, J.; VIKNER, S. (eds.) *Optimality-theoretic syntax*. Cambridge: MIT Press, 2001.
13. McCARTHY, J.; PRINCE, A. *Prosodic Morphology I: constraint interaction and satisfaction*. Manuscript, University of Massachusetts, Amherst, and Rutgers University, New Brunswick, N. J., 1993.
14. PRINCE, A.; SMOLENSKY, P. *Optimality Theory: constraint interaction in Generative Grammar*. RuCCs Technical Report 2, Rutgers University Center for Cognitive Science, Piscataway, N. J., 1993.
15. SELLS, P. (ed.) *Formal and empirical issues in optimality theoretic syntax*. Stanford: CSLI Publications, 2001.